SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

R. Goethika

Programme: **B.E / B.Tech**
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Max Marks: **50 Marks**
Date : _____

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

X \ Y	Y = 0	Y = 1
X = 2	0.20	0.10

Analyse the distribution by (a) finding the marginals of X and Y, (b) computing E[X], Var(X) and E[Y], (c) computing $P(X = 2 \mid Y = 1)$, (d) computing Cov(X, Y) and the correlation coefficient, and (e) deciding whether X and Y are independent, with a justification.

(6) (CO1) (BL4)

Task 4. A temperature sensor output is modelled as $X \sim N(\mu = 50, \sigma^2 = 16)$.

(a) Calculate $P(X > 58)$ and $P(46 < X < 54)$.

(b) A second sensor gives the readings 12, 15, 11, 14, 13. Derive the maximum-likelihood estimates of its mean and variance and compare the ML variance with the unbiased estimate, stating which is biased and why. (Use $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.)

(4) (CO1) (BL4)

Task 5. A source emits four symbols with probabilities (1/2, 1/4, 1/8, 1/8). Calculate its entropy, compare it with the entropy of a uniform source over four symbols, and analyse the redundancy. Also compute the average length of the code {0, 10, 110, 111} and comment on its optimality.

(3) (CO1) (BL4)

Task 6. A model predicts class probabilities Q = (1/3, 1/3, 1/3) while the true distribution is P = (0.5, 0.25, 0.25). Compute KL(P‖Q) and KL(Q‖P) in bits, and analyse what the asymmetry implies when KL divergence is used as a training loss.

(3) (CO1) (BL4)

Task 7. Two binary variables A and B have the joint distribution $P(0,0) = 0.4$, $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$. Calculate H(A), H(B), the joint entropy H(A,B), the conditional entropy H(A|B) and the mutual information I(A;B). Analyse what the value of I(A;B) says about using B as a feature to predict A.

(4) (CO1) (BL4)

Task 8. A one-dimensional feature x follows $p(x|\omega_1) = N(2, 4)$ for class ω_1 and $p(x|\omega_2) = N(6, 4)$ for class ω_2, with priors $P(\omega_1) = 0.6$ and $P(\omega_2) = 0.4$ and zero–one loss. Derive the Bayes decision threshold x^*, classify $x = 4.5$, and calculate the overall probability of error. Analyse how the threshold moves as the prior of ω_1 increases.

(4) (CO1) (BL4)

Task 9. For an observation x, the posteriors are $P(\omega_1|x) = 0.3$ (disease) and $P(\omega_2|x) = 0.7$ (healthy). Action α_1 = 'declare disease', α_2 = 'declare healthy'. The loss matrix is $\lambda(\alpha_1|\omega_1) = 0$, $\lambda(\alpha_1|\omega_2) = 8$ (unnecessary treatment), $\lambda(\alpha_2|\omega_1) = 50$ (missed disease), $\lambda(\alpha_2|\omega_2) = 0$. Evaluate the conditional risks, decide the minimum-risk action, compare it with the MAP decision and determine the posterior $P(\omega_1|x)$ above which α_1 is optimal.

(3) (CO1) (BL5)

Task 10. Polynomial regression models of increasing degree were fitted to the same dataset, giving the mean-squared errors below.

Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1

Degree	1	2	3	5	9
Validation MSE	9.8	4.6	2.9	3.6	11.8

Calculate the generalisation gap for each degree, diagnose under-fitting/over-fitting using the bias–variance trade-off, select the best model with justification, and recommend two techniques to improve the degree-9 model.

(4) (CO1) (BL5)

Task 11. A coin is tossed 10 times and shows 7 heads. The bias θ = P(head) is given a Beta(2, 2) prior. Calculate the ML estimate, the MAP estimate and the posterior mean of θ. Then evaluate how the three estimates change if the data were 70 heads in 100 tosses, and explain the role of the prior.

(4) (CO1) (BL5)

Task 12. A hospital wants an early-warning system that raises an alert when an ICU patient is likely to develop sepsis within the next 6 hours. Historical electronic health records (vital signs, lab values, outcomes) are available.

(a) Design the learning system: state the task T, performance measure P and experience E, the target function, a suitable input representation, a learning algorithm and the validation strategy. [3]

(b) On a validation cohort of 1,000 patients, 5% develop sepsis. The trained model has sensitivity 0.90 and specificity 0.80. Compute TP, FN, FP and TN. [1]

(c) Compute the positive predictive value (PPV) and negative predictive value (NPV). [1]

(d) A missed sepsis case costs ₹100 (unit cost) and a false alarm costs ₹5. Compare the total cost of (i) using the model, (ii) alerting every patient and (iii) never alerting. [2]

(e) Recommend a deployment policy and justify how the alert threshold should be set using Bayes decision theory. [1]

(8) (CO1) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - 1

Programme	B.E / B.Tech	Course Code	MLA02
Course Name	Fundamentals of Machine learning.	Date	
Duration	1 Hr 30 Min	Max Marks	50

Register No.: 192485101

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	6
Task 4	4	4
Task 5	3	3
Task 6	3	3
Task 7	4	4
Task 8	4	3
Task 9	3	3
Task 10	4	4
Task 11	4	4
Task 12	8	7
TOTAL	**50**	48

Total in words:

Assignment-1

MLA0204 - Fundamentals of Machine learning

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Reg NO:- 19242510 1

Q1) Task1 - Identify the types of Machine learning

1. loan Default prediction - supervised learning
The Model is trained using previous customer records labelled as default or non-default. It predicts whether a new customer is likely to default on a loan.

2. customer segmentation - Unsupervised learning
customers are grouped Based on similarities in their Purchasing Behaviour, income and interests No predefined labels are required

3. Credit spend prediction - supervised learning
The Model uses past customer data to predict a continuous Numerical value, such as the Amount a customer may spend.

4. ATM cash refill - Reinforcement learning
The system learns an effective cash refilling strategy by considering rewards and penalties Associated with cash Availability and refill costs.

5) Fraud detection - semi supervised learning

Semi supervised learning - The Model uses a small amount of labelled transcation data along with a large amount of Unlabelled data to identify suspicious transactions.

Q3) Task 2 - Application of Baye's theorem

Baye's theorem is used to calculate the probability of an event based on previously available the information.

The formula is

$$P(D|+) = \frac{P(+|D) \times P(D)}{P(+)}$$

Where;

$P(D|+)$ = probability of having the disease given a positive test

$P(+|D)$ = probability of a positive result when the person has the disease.

$P(D)$ = probability of having the disease

$P(+)$ = Overall probability of a positive test result.

Substituting the given values

$$P(D|+) = 0.1624$$

Converting into percentage

$$P(D|+) = 0.1624 \times 100$$

$$= 16.24\%$$

Q2)

Task 3 - Joint probability distribution

The joint probability distribution represents the probability of two random variables occurring together.

1. Marginal probability of x :-

The Marginal probability of x is obtained by adding the joint probabilities corresponding to each value of x.

$P(x) = (0.30, 0.40, 0.30)$

2. Marginal probability of y :-

Similarly adding the joint probabilities for each value of y.

$P(y) = (0.45, 0.55)$

3. Expected Value :-

The Expected value is calculating using

$$E(x) = \sum x p(x)$$

$$\boxed{E(x) = 1}$$

for y:

$$E(y) = \sum y p(y)$$

$$E(y) = 0.55.$$

4. Variance of x :-

Variance measures how much the values differ from their Mean

$$Var(X) = E(X^2) - [E(X)]^2$$

$$Var(X) = 1.60 - (1)^2$$

$$\boxed{Var(X) = 0.60}$$

5) conditional probability

The conditional probability is

$$P(X=2 \mid Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)}$$

$$P(X=2 \mid Y=1) = 0.1818$$

6) Covariance and Correlation

Covariance Measures the relationship Between two variables.

$$Cov(X,Y) = E(XY) = E(X)\,E(Y)$$

$$Cov(X,Y) = -0.10$$

The correlation coefficient is

$$r = -0.2593$$

Task 4 – Normal Distribution and Variance

Part A – Normal Distribution

Sol

Given

$$X \sim N(50, 16)$$

Mean

$$\mu = 50$$

Variance $\sigma^2 = 16$

Standard deviation $\sigma = \sqrt{16} = 4$

1. Calculate $P(x > 58)$

Convert x into the standard Normal variable z

$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{58 - 50}{4}$$

$$= 2$$

From the standard Normal table

$$P(z > 2) = 0.0228$$

$$P(x > 58) = 0.0228$$

2. Calculate $P(46 < X < 54)$

For the lower limit

$$z_1 = \frac{46 - 50}{4} = -1$$

For the upper limit

$$z_2 = \frac{54 - 50}{4} = 1$$

$$\therefore P(46 < X < 54) = P(-1 < z < 1)$$

$$P(46 < X < 54) = 0.6826$$

Part B:- Mean and Variance

Given data

12, 15, 11, 14, 13

Mean

12, 15, 11, 14, 13

$$\bar{x} = \frac{\sum x}{n}$$

$$= \frac{12+15+11+14+13}{5}$$

$$\bar{x} = 13$$

Calculate squared deviations:

x	$x - \bar{x}$	$(x-\bar{x})^2$
12	−1	1
15	2	4
11	−2	4
14	1	1
13	0	0
Total		10

Maximum likelihood Estimate (MLE)

variance

$$\sigma^2 = \frac{\sum (x-\bar{x})^2}{n}$$

$$= 10/5 = 2$$

Unbiased sample variance

$$s^2 = \frac{\sum (x-\bar{x})^2}{n-1}$$

$$= 10/4 = 2.5$$

Tasks - Entropy and Redundancy

Entropy Measures the Uncertainty or information content of a random variable.

The Entropy Formula is

$$H(x) = -\sum p(x) \log_2 p(x)$$

Using the given probabilities

$$H(x) = 1.75 \text{ bits}$$

For four Equally probable symbols, Maximum Entropy is

$$H_{max} = \log_2 4$$

$$H_{max} = 2 \text{ bits}$$

Calculation of Redundancy:

$$R = H_{max} - H(X)$$

$$R = 2 - 1.75$$

$$R = 0.25 \text{ bites}$$

2+1

Percentage redundancy

$$R\% = \frac{0.25}{2} \times 100$$

$$R = 12.5\%$$

The Average code length is

$$L = 1.75 \text{ bits}$$

∴ Thus, the entropy is 1.75 bits and the redundancy is 12.5%.

Q6) Kullback Leibler Divergence

KL Divergence Measures the difference the Between two probability distributions. It indicates how Much information is lost when one distribution is used to represent another.

the formula is

$$D_{KL}(P||Q) = \sum_i P(i) \log_2 \frac{P(P)}{Q(i)}$$

Substituting the given probability values

$$D_{KL}(P||Q) = 0.8496$$

$$D_{KL}(P||Q) = 0.8496 \text{ bits.}$$

Reversing the distributions

$$D_{KL}(Q||P) = \sum_i Q(i) \log_2 \frac{Q(i)}{P(i)}$$

$$D_{KL}(Q||P) = 0.08170 \text{ bits.}$$

∴ the two values are different Because KL divergence is not symmetric.

Q7) Entropy and Mutual Information

Entropy Measures the Uncertainity in individual random variables, while Joint Entropy Measures the Uncertainity in two variables together.

1. Entropy of A

$$H(A) = -\sum P(A) \log_2 P(A)$$

$H(A) = 1 \text{ bit}$

2. Entropy of B

$$H(B) = -\sum P(B) \log_2 P(B)$$

$$\boxed{H(B) = 1 \text{ bit}}$$

3. Joint Entropy

$$H(A,B) = -\sum_a \sum_b P(a,b) \log_2 P(a,b)$$

$$H(A,B) = 1.7219 \text{ bits}$$

4. Conditional entropy

Conditional Entropy represents the Remaining Uncertainity in A when B is known

$$H(A/B) = H(A,B) - H(B)$$

$$= 1.7219 - 1$$

$$H(A|B) = 0.7219 \text{ bits.}$$

5. Mutual information

Mutual information Measures how Much information one variable provides about another.

$$I(A:B) = H(A) + H(B) - H(A,B)$$

$$= 1 + 1 - 1.7219$$

$$= 2 - 1.7219$$

$$\boxed{I(A,B) = 0.2781 \text{ bites}}$$

Q2) Task 8 – Gaussian Bayes classifier

A Gaussian Bayes classifier uses probability distributions to assign an observation to the Most likely class.

The decision boundary obtained for the given distributions is

$$x^* = 4 + \ln(1.5)$$

$$\boxed{x^* = 4.4055}$$

3

classification of the given value

Given

$$x = 4.5$$

Comparing with the decision Boundary

$$4.5 > 4.4055$$

Therefore, the observation belongs to class:

$$\boxed{w_2}$$

The classification error probability is

$$\boxed{15.384.}$$

∴ The decision threshold is 4.4055 and the given observation $x = 4.5$ classified as w_2.

Q3) Task 9 – Minimum Risk classification

Minimum risk classification selects the decision that products the lowest expected loss.

The risk of choosing Disease is $\boxed{R(D) = 5.6}$

The risk of choosing Number 2 is

$$P(\text{Fish}) = 15$$

Comparing the two risks

$$R(2) < R(\text{Fish})$$

$$5 < 15$$

Choosing 2 leads to the lower expected risk

Decision threshold

3

$$P(D|2) > \frac{C_{FP}}{C_{FP} + C_{FN}}$$

Substituting the given costs

$$P(D|x) > \frac{2}{2+50}$$

$$P(D|x) > 0.1279$$

Threshold = 13.79 %.

Q10) Task 10 - Generalization and Overfitting

Generalization refers to how well a Machine learning Model performs on new, Unseen data Overfitting occurs when a Model performs Very well on training data But Poorly On test data.

Generalization Gap = Test Error - Training Error

the calculated results are

Polynomial Degree	Generalization Gap
1	0.5
2	0.5
3	0.7
5	2.7
9	11.7

As polynomial degree increases, the Generalization gap becomes larger, particularly for degrees 5 and 9. this coefficient indicates increasing overfitting.

Polynomial Degree = 3

Methods to reduce Overfitting

1. Regularization – Add a penalty to reduce Model complexity.
2. Feature selection – Removes Unnecessary input variables.
3. Cross validation – Tests Model Performance On different data subsets.
4. Reducing Model Complexity – Avoids using an Unnecessarily Complex model.

11)

Task II – Bayesian Estimation of Coin Tosses

Given a Beta prior distribution

$$\text{Beta}(\alpha, \beta) = \text{Beta}(2, 2)$$

$$\text{Beta}(\alpha + h, \beta + T)$$

Part A: 7 Heads out of 10 tosses

Number of Heads $= 7$

Number of tails $= 3$

1. **Maximum Likelihood Estimate**

$$\hat{p}_{MLE} = h/n$$

$$= 7/10$$

$$\boxed{MLE = 0.7000}$$

2. **Maximum A posteriori Estimate**

$$\hat{p}_{MAP} = \frac{\alpha + h - 1}{\alpha + \beta + n - 2}$$

$$= \frac{2 + 7 - 1}{2 + 2 + 10 - 2}$$

$$\boxed{MAP = 0.6667}$$

3. **Posterior Mean**

$$E(p) = \frac{\alpha + h}{\alpha + \beta + n} = \frac{2 + 7}{2 + 2 + h} = 0.6429$$

Part B:- 70 heads out of 100 tosses

Number of heads = 70

Number of tails = 30

(i) Maximum Likelihood Estimate

$$MLE = \frac{70}{100}$$

$$\boxed{MLE = 0.7000}$$

(ii) Maximum A Posteriori Estimate

$$MAP = \frac{2 + 70 - 1}{2 + 2 + 100 - 2}$$

$$\boxed{MAP = 0.6961}$$

(iii) Posterior Mean

$$E(p) = \frac{2 + 70}{2 + 2 + 100}$$

$$\boxed{E(p) = 0.6963}$$

Q1y Task 19 :- sepsis prediction and cost Analysis

Part A:- confusion Matrix

A confusion Matrix Compares actual results with Predicted Matrix.

Actual / Predicted	positive	Negative
Positive	45 (TP)	5 (FN)
Negative	190 (FP)	760 (TN)

Where,

TP = True positive

TN = True Negative

FP = false positive

FN = false Negative.

Part B :- Performance Measures

1. precision

Precision Measures the proportion of positive predictions that are actually positive.

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$= \frac{45}{45+190}$$

$$= \frac{45}{235}$$

Precision = 19.15 %.

2. Negative predictive value

$$NPV = \frac{TN}{TN+FN}$$

$$= \frac{760}{760+5}$$

NPV = 99.35 %.

Part c:- cost analysis

1450 < 4750 < 5000.

Part D:-

$$P(D|x) \Rightarrow \frac{C_{FP}}{C_{FP} + C_{FN}}$$

Substituting the given costs

$$P(D|x) > \frac{5}{5+100}$$

$$P(D|x) > 0.0476$$

Decision threshold > 0.0476.